A Systematic Literature Review of Machine Learning Approaches for Migrating Monolithic Systems to Microservices

Source: I. Trabelsi *et al.* "A Systematic Literature Review of Machine Learning Approaches for Migrating Monolithic Systems to Microservices," *IEEE Transactions on Software Engineering*, 2025. [Online]. Available: https://ieeexplore.ieee.org/document/11145241/

Introduction (What is the Paper about?)

- **Research GAP:** No prior work has yet systematically investigated *Machine Learning (ML)* Approaches during the migration from monolithic systems to microservices
- This paper present a *Systematic-Literature Review (SLR)* to address the above gap

Methods

- Created an *Systematic-Literature Review (SLR)*
  - followed the updated *Preferred Reporting Items for Systematic Review and Meta-Analysis (PRISMA)* statement for reporting systematic reviews
- Screened 2,301 potentially relevant studies from eight digital libraries considering publications between 2015 and 2024
  - used Inclusion- and Exclusion criteria
  - used *Snowball Sampling*
  - retained 81 studies for analysis after quality assessment

Scientific Benefit

- Comprehensive **Understanding of how ML** can be used to support the Migration
  - systematic analysis of the automated migration phases using ML
    - identifying gaps
  - synthesis of the types of inputs used in ML-driven approaches
    - including granularity, sources and how it is preprocessed
  - analysis of the applied ML techniques
    - highlighting commonly used models, emerging techniques
  - exploration of evaluation practices identifying common metrics, benchmarks and success criteria
  - discussion of the challenges encountered including scalability, data availability and the interpretability of models
  - Set of recommendations for practitioners and researcher
- **Phases are differential well studied**
  - Well studied phases: *Monitoring, Deployment* and *Service Identification*
  - Less well studied: *Pre-Migration*, *Packaging Microservices, Generating necessary code for microservice APIs* and *Implementing Design Patterns*
  - *Automating code generation* and *Packaging Tasks* has received limited attention despite the potential of emerging technologies like LLM
- **Extracted Key Challenges** hindering adoption in practical scenarios:
  - Insufficient availability of high-quality data
  - Scalability and complexity concerns
  - Insufficient tool support
  - Absence of standardized benchmarks, datasets and baselines

Challenges / Future Work

- Automation of *Business Processes (BP)* discovery and refactoring remains largely unexplored

  - valuable opportunity for future research using *NLP*, *Process Mining* or *LLMs*
- Scarcity of real-world datasets
  - raises concerns about the practical applicability of ML techniques
- Evaluation Practices vary widely
  - System adaptability and real-world validation is often overlooked
- Enhancing data accessibility through industry collaboration, privacy-preserving data-sharing frameworks and development of benchmark datasets will be crucial
- Exploring hybrid ML approaches integrating multiple learning paradigms could be beneficial to accuracy and adaptability in the migration process

Limitations

Paper Outline and Read Progress

1. [X] Introduction
2. [X] Background and Related Work on monolithic to microservice migration
3. [ ] Description of the SLR Methodology and used analysis processes
4. [ ] Statistical Overview of the selected Literature
5. [ ] Current Phases of migration being currently automated using ML
6. [ ] Types of data collected and processes using ML
7. [ ] ML techniques applied in migration
8. [ ] How ML-based techniques are evaluated
9. [ ] Challenges and Limitations of applying ML
10. [ ] Observations and Recommendations
11. [ ] Limitations
12. [X] Conclusion

Own Comments

- (+) Key Paper which might give an overview over the current field
- (-) Introduction feels a bit repetitive. It could be shortened
- (-) [15] citation is used twice for two different sources